

David Rey

Unity & Unreal Developer | Technical Artist | Gameplay Programmer

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Portfolio: viid21.github.io/UnityDev_Portfolio/

GitHub: github.com/Viid21

Professional Summary

Unity & Unreal Developer and Technical Artist with 3 years of experience shipping PC, mobile, VR, and AR projects. Strong foundation in C# programming, gameplay systems, shader development, and real-time performance optimization across both engines. Proficient in Unity tooling including Shader Graph, Scriptable Objects, FSMs, Profiler, and Frame Debugger, as well as 2D and 3D pipelines. Experienced in Unreal Engine for VR development, with deep knowledge of draw call optimization and engine debugging workflows. Familiar with CI/CD pipelines using Git and Unity Cloud Build, asset management best practices, and version control in team environments. Committed to delivering production-quality work with a focus on performance, visual quality, and clean technical execution.

Core Skills

Unity · Unreal Engine · C# · Gameplay Programming · Technical Art · Shader Graph · Performance Optimization · Profiler · Frame Debugger · Scriptable Objects · Finite State Machines (FSM) · VR Development · AR Development · Real-Time Rendering · 2D Development · Git · CI/CD · Unity Cloud Build · Android Builds

Professional Experience

Eonagate

VR Developer | Performance Engineer

April 2026 – Present | Remote

- Audited and profiled a complex 3D scene for VR deployment, identifying GPU bottlenecks and performance issues using Unreal's built-in debugging and profiling tools.
- Reduced draw calls significantly to meet VR's strict framerate requirements, resolving GPU-side bottlenecks and ensuring smooth real-time performance.
- Implemented full VR interaction systems from scratch, covering all player-scene interactions within Unreal Engine.

- Baked lighting and shadows to optimize real-time rendering cost while preserving the photorealistic visual quality required by the project.
- Coordinated with the 3D art team, providing technical direction on asset and texture optimization to meet VR performance budgets.

Arcadio Lab

Unity Programmer | Technical Artist

July 2024 – September 2025 | Barcelona

- Developed and maintained gameplay systems in Unity within a multidisciplinary team following Scrum methodologies.
- Implemented and iterated on gameplay features in an active production environment, resolving bugs and ensuring stability across builds.
- Created shaders and visual effects using Shader Graph, balancing visual quality and runtime performance.
- Optimized scenes using Unity Profiler and Frame Debugger to identify bottlenecks and improve real-time performance and build stability.
- Collaborated closely with Game Designers and 3D Artists to define and prioritize content for the demo, contributing to creative decisions and ensuring technical feasibility.

Forged Horizon

Unreal Engine VR Developer | Technical Artist

September 2023 – June 2024 | Barcelona

- Designed and implemented VR interaction mechanics and user-facing gameplay systems using Blueprints in Unreal Engine
- Configured realistic lighting setups to enhance immersion and visual fidelity in VR environments
- Optimized materials and shaders using instancing techniques to reduce draw calls and meet VR framerate requirements
- Applied technical art workflows to balance visual quality and performance across different VR devices and platforms
- Awarded Best Project of the Year at Deia Escola d'Art i Disseny (2024).

Grupo 3 Studio

Unity Programmer | Technical Artist

September 2022 – July 2023 | Barcelona

- Designed and implemented mission and dialogue systems using Scriptable Objects for scalability and maintainability
- Developed a player controller with Unity's New Input System and FSM architecture to manage player actions and animations

- Improved gameplay architecture to reduce bugs, speed up iteration, and support collaboration between designers and developers
- Awarded Best Project of the Year at Deia Escola d'Art i Disseny (2023).

Selected Projects

I.P.O.D. Atrapamonstres

Unity AR Developer

2025

- Developed and integrated AR features in Unity using Vuforia for a live, published mobile project.
- Implemented gameplay features and fixed issues in an active production environment.
- Prepared, tested, and maintained Android builds across different devices.
- Created VFX and visual feedback to improve clarity and responsiveness of player interactions.
- Used Unity Cloud services and SQL-based data workflows to manage player data, persistence, and tracking.
- Balanced visual quality with mobile performance constraints to improve stability and usability on real devices.

Golden Hand

Unreal VR Developer

2024 – 2025

- Developed VR mechanics in Blueprints, including climbing, archery, and escape-room-style puzzles.
- Optimized performance for Meta Quest 3.
- Set up realistic lighting and material instancing for immersive VR environments.

A non farming game

Unity Developer

2023 – 2024

- Built mission and dialogue systems using Scriptable Objects.
- Developed a player controller using Unity's New Input System and FSM architecture.
- Implemented enemy AI and boss fight mechanics.
- Optimized graphics using LODs, Occlusion Culling, and Frustum Culling.
- Created a dynamic day/night cycle system.

Education

IT Academy de Barcelona Activa

Specialization Course in Java Spring Boot Development

September 2025 – March 2026

Deià Escola d'Art i Disseny

Higher National Diploma (HND) in Audiovisual Graphics – Virtual Reality and Gamification

September 2023 – June 2024

Deià Escola d'Art i Disseny

Higher National Diploma (HND) in Audiovisual Graphics – Video Games and Virtual Environments

September 2020 – June 2023

Languages

Catalan: Native

Spanish: Native

English: Advanced